Pencil Pressure Safety Lab

Spread a force before it dents the target.

Win condition: Largest area protected with clear data.

THE SCIENCE YOU ARE USING

Pressure vs force. Pressure is force divided by area. The same push on a sharp point dents a surface, but spread over a wide area it does not. That is why a sharp pencil pierces and a flat board does not: same force, different area.

Spreading stress. Safety design often means spreading a load so no single spot takes too much. You design a spreader that protects foam from a standard load and show the protected area with data.

HOW TO BUILD AND RUN IT

01 **Test the bare point.** Press a standard load through a pencil point into foam. Measure the dent.

02 **Design a spreader.** Build a cardboard spreader that distributes the same load over more area.

03 **Apply the standard load.** Press the same weight through your spreader onto fresh foam.

04 **Measure protected area.** Compare the dent. Less damage over more area means lower pressure.

05 **Redesign for more spread.** Improve the spreader to protect a larger area and re-test.

Safety: No skin tests. Use foam only. Allergy: None.

MATERIALS

Pencils	shared
Foam	per team
Cardboard	shared
Weights	shared
Graph paper	per team

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why): ______

The ONE variable we are testing: ______

Baseline result: ______

After redesign: ______

DEFEND YOUR DESIGN

What evidence made you change your design? ______

If you ran it again, what is the ONE thing you would change? ______

HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Pressure spread	40
Explanation	25
Data quality	20
Redesign	15
Total	**100**